Joel Jacob

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Education

Carnegie Mellon University – BS in Mechanical Engineering — Expected May 2028

Skills

Design & Analysis:	SolidWorks, Onshape, ANSYS (Discovery, Fluent)
Controls & Programming:	Python, MATLAB, Java, C++, Arduino, Raspberry Pi, Git
Manufacturing & Fabrication:	3D Printing, Machining, Laser Cutting, Soldering, Assembly

Projects

NASA Lunabotics Challenge — Nov 2025 – Present

- Designed and fabricated excavation subsystem components as a member of CMU Moon Miners, winning the best presentation by a rookie team at the 2026 NASA Lunabotics Competition
- Owned the dual-channel bucket excavator design through design reviews, 3D printed prototyping, and machined final version that outperformed the baseline in terms of regolith throughput
- Minimized weight through structural FEA in ANSYS Discovery while maintaining required safety factors

National Havoc Robot League — Sept 2024 – Present

- Founded a new robot project under CMU Combat Robotics, coordinating a team to design, develop, and manufacture a 3 lb undercutter combat robot to compete in the National Havoc Robot League
- Designed 10+ custom robot components on Onshape, including a TPU chassis, urethane wheels, and AR500 steel spinner weapon, accelerating iteration before 3D printing, rubber casting, and machining
- Optimized ideal motor Kv and weapon moments of inertia using MATLAB analysis to maximize impact effectiveness while retaining stable rotational performance

FIRST Robotics Competition — Oct 2021 – June 2024

- Oversaw the 12-member control department as a lead of FRC Team 5123 to network and program a complete CAN bus-based controller system on the robot, which won the 2023 FRC NYC Regional Championship
- Improved autonomous driving accuracy and reduced steady-state error by 65% by implementing vision-based localization with Raspberry Pi and tuning PID gains
- Programmed Java command-based robot code with custom teleoperated controls and tested it in robotic simulation programs, improving driving precision and on-field responsiveness during competition

Experience

Research Assistant, Carnegie Mellon University – Pittsburgh, PA — Sept 2025 – Present

- Contributed to the development of a wearable assistive device as a member of the MetaMobility Lab at CMU by integrating sensing and deep learning to improve human mobility support
- Redesigned hip exoskeleton hardware for improved user fit by applying human biomechanics principles, leveraging SolidWorks and 3D printing to achieve precise motor-joint alignment and a 30% weight reduction

Retail Sales Associate, S & A Stores Inc – Bronx, NY — June 2025 – Aug 2025

- Maintained organization of products across high-traffic areas, integrating loss prevention protocols, to improve overall customer experience, comply with presentation standards, and reduce product stock-outs by 20%

Green Team Associate, Groundwork Hudson Valley – Yonkers, NY — June 2022 – June 2024

- Reduced harmful sources against native biota by taking part in environmental engineering at over 20 job sites